

Advanced DataBases Tuto # 2 : Optimisation

Exercise 01 :

Consider the following relational schema:

Owner (OwnerNum, first name, last name, address, phone number).

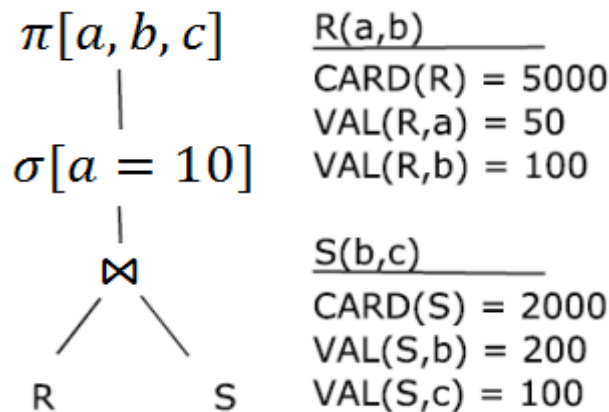
Property (PropertyNum, city, type, rental_price, OwnerNum*)

Customer (CustomerNum, firstnameC, lastnameC, addressC, preftype, BudgetMax)

Visit (PropertyNum*, CustomerNum*, VisitDate)

- Express the following query in SQL: on behalf of tenants, who have visited apartments, find all the accommodations which correspond to their requirements and which belong to the owner with number CP93.
- Give the algebraic tree that corresponds to the SQL query.
- Optimize the query using transformation rules.

Exercise 02 : Consider the following algebraic tree as well as the statistical information related to the manipulated tables.



Indicate the possible executions and evaluate their execution cost.

Exercise 03 :

Given the following relational schema and SQL query:

Student (idS, name, age, address)

Book (idB, title, author)

Borrow (idS*, idB*, date)

Query : SELECT s.nom
FROM Student s, Book b, Borrow br
WHERE s.idS = br.idS
AND b.idB = br.idB
AND b. author = 'DATE'
AND s.age=20

And assuming that:

- There are 10,000 records of the table student stored on 1,000 pages.

- There are 50,000 records of the table Book stored on 5,000 pages.
- There are 300,000 records of the table borrow stored on 15,000 pages.
- There are 500 different authors.
- Student ages range from 10 to 24 years old.

Questions :

- Give an execution plan for this query, assuming that there are no indexes and that the data is not sorted on any attribute.
- Calculate the cost of this plan and the cardinality of the result.
- Suggest two indexes and another plan for this query.
- Calculate the cost of the new plan.

Additional exercise 04 :

Let us consider a database consisting of the following relations:

Production (NumP, Type, Model, Qty, Machine)

DetailOrder (NumO*, NumP*)

Order(NumO, Client, Seller, Amount)

And consider the following profiles:

CARD(Production) = 200 000	VAL(NumP) = 200 000
CARD(DetailOrder) = 50 000	VAL(Type) = 4
CARD(Order) = 10 000	VAL(Model) = 400
SIZE(NumP) = 10	VAL(Qty) = 100
SIZE(Type) = 1	VAL(Machine) = 50
SIZE(Model) = 10	VAL(NumO) = 10 000
SIZE(Qty) = 10	VAL(Client) = 400
SIZE(Machine) = 10	VAL(Amount) = 5000
SIZE(NumO) = 5	VAL(Seller) = 25
SIZE(Client) = 30	
SIZE(Amount) = 10	
SIZE(Order) = 45	
SIZE(DetailOrder) = 15	
SIZE(Production) = 41	
SIZE(Seller) = 20	
CARD(X) : cardinality of the table X i.e. the number of tuples in the table X.	
SIZE(Y) : size of the attribute Y in bytes.	
VAL(Z) : the number of distinct values of the attribut Z.	

Describe the algebraic optimization and the calculation of profiles on intermediate results for the following queries. To this end, write them into SQL then translate them into algebraic language. For the queries b and c, which require join operations, give the order that seems the most appropriate one.

Queries:

- Give the available quantity of the product R2778.
- Give the machines were necessary for the production of the products purchased by Mr. Ahmed.
- Find the clients that had purchased a product with the number P24 from the seller Smith.